

## EDUCATION

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Texas A&M University, College Station, Texas

Expected Graduation: May 2028

Bachelor of Science in Interdisciplinary Engineering

GPA: 3.71

Focus: Biomedical Engineering and Computer Science

- **Engineering Honors** (Top 10% of Engineering cohort)
- **Scholarships:** Timothy J. Holloman Engineering Scholarship; Multidisciplinary Engineering Departmental Scholarship; J. Malon Southerland Aggie Leader Scholarship; Cary N. Smith '34 Memorial Scholarship

## EXPERIENCE

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### Tabor Research Group

May 2025 – Present

Undergraduate Researcher

- Developed a Python and RDKit-based computational screening pipeline to screen **1+ billion molecules** with **SMARTS-based structural screening**, reducing manual effort by **over 90%** and collecting valid structural candidates for downstream analysis
- **Automated** conformer generation, geometry optimization, and energy analysis for **100,000+ photoacid candidates** across **6 molecular forms** using RDKit, CREST, AIMNet2, Gaussian 16, SLURM, and **high-performance computing systems** including Texas A&M's **Grace** and **VISION (NVIDIA)**
- Analyzed **5+ candidate-selection metrics**, including relative energies, heavy atom counts, closed-shift geometries, and pKa predictions using **pretrained machine learning models** to prioritize candidates for downstream TD-DFT
- Delivered **5+** slide-based research presentations to a lab group of **15+** graduate students, postdocs, and Professor Tabor, effectively communicating screening results and candidate selection

### Society of Asian Scientists and Engineers (SASE)

August 2024 – Present

Science Director

- Elected Science Director for TAMU's SASE chapter, sponsored by companies such as Chevron, Boeing, Capital One, and Lockheed Martin, to oversee the planning of **5+ science-related events** per semester for **200+ members**
- Managed an allocated budget and lead a **14-member** committee to organize engaging events including research workshops, outreach events, and academic fair representation to promote undergraduate research involvement

## PROJECTS

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### Jung Chance Fine Art Website

June 2026 – July 2026

- Developed a **responsive** fine art website across **5+ pages** using Next.js, TypeScript, Tailwind CSS, and Sanity CMS, eliminating a **\$300 annual** website subscription
- Built reusable gallery, event, contact, and content-management features, enabling the client to update artwork independently **without modifying code**

### Strive (ACC)

September 2024 – May 2025

Project Member

- Collaborated with a 17-member team to develop Strive, a mobile app for class scheduling built with JavaScript, React Native, and Firebase
- Contributed to frontend development, supported UI design through Figma, and provided usability feedback to enhance app functionality and user experience

## TECHNICAL SKILLS

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**Languages:** Python, C++, HTML, JavaScript, CSS, SQL, SMILES/SMARTS

**Frameworks & Libraries:** React, Next.js, React Native, Tailwind CSS, RDKit, pandas, NumPy, SciPy, matplotlib, PyTorch

**Software:** Gaussian 16, CREST, ChemDraw, SOLIDWORKS, AutoCAD 2D/3D, Onshape, Figma, Excel

**Developer Tools:** Git/GitHub, VS Code, PyCharm, Sanity CMS, SLURM, High-Performance Computing

**Relevant Coursework:** Data Structures & Algorithms, Biomedical Computing & Design, Program Design & Concepts, Discrete Structures for Computing, Biostatistics & Data Visualization, Organic Chemistry I & II, Physiology for Bioengineers